POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2019
Programme: BE | Full Marks: 100
Course: Physics | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why compound pendulum is preferred than simple pendulum. For compound pendulum, prove that the minimum time period is obtained if the point of suspension and point of oscillations are equidistant from centre of gravity. 1+8

   b) What is amplitude, the wave length and the velocity of the wave represented by y=5 $\sin(6\pi t + 4x)$ Where distance and time are measured in S.I Units? 6

2. a) What are the differences between Interference and diffraction of light? Explain the fraunhoffer diffraction through single slit and hence calculate the width of central maximum. 1+8

   b) A classroom has dimension of 20x15x5 m^3. The reverberation time is 3.5sec. Calculate the total absorption of its surface and the average absorption coefficient. 6

3. a) Define electric dipole. Develop a relation of electric field intensity of a dipole not lying along equatorial line. 1+8

   b) A soap film of 5×10^{-7} m thick is viewed at an angle of 35^0 to the normal. Find the wavelengths of light in the visible spectrum which will bw absent from the reflected light, given that the refractive index of the film is 1.33. 6

4. a) Derive an expression for the growth and decay of current in L-R circuit. Also, show that the value of transient current never exceed the steady current. 8+1

   b) Find the magnitude of induced emf in a 200 turns coil with a cross-sectional area of $0.16m^2$, if the magnetic field through the coil changes from $0.10wm^{-2}$ to $0.50\ wm^{-2}$ at the uniform rate over a period 0.02 sec. 6

5. a) Write Maxwell's equations in integral and differential form. Using Maxwell's equation in free space show that speed of electromagnetic wave is equals to the speed of light. 2+7

b) Obtain the charging time constant of a capacitor in a RC circuit such that current through the resistor is decreased by 50% of its peak value in 10 seconds. 6

6. a) What is wave function? Describe its significance. Derive Schrodinger time independent wave equation for a free particle like electron. 2+7

b) Prove that the equation of continuity $\vec{\Delta}.\vec{J} + \frac{\partial \rho}{\partial t} = 0$ 6

7. Write short notes on: (**Any two**) 2×5

a) Describe Spontaneous emission and Stimulated emission of a radiation

b) Resistance and Resistivity

c) Semi-conductor dopping